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(54) **LED LIGHTING SYSTEMS WITH TRIAC DIMMERS AND METHODS THEREOF**

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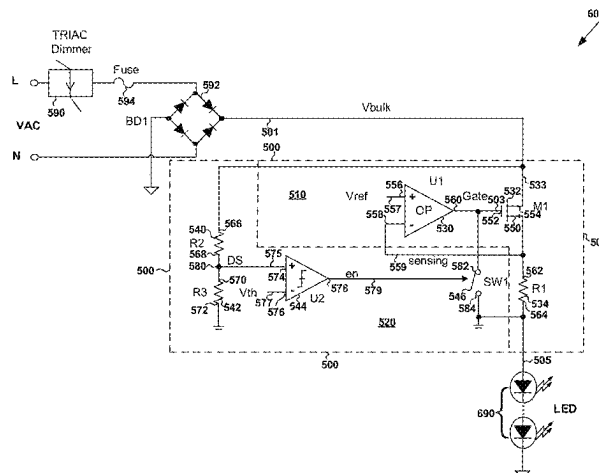
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(57) **ABSTRACT**
System and method for controlling one or more light emitting diodes. For example, the system for controlling one or more light emitting diodes includes a current regulation circuit coupled to a cathode of one or more light emitting diodes. The one or more light emitting diodes include the cathode and an anode configured to receive a rectified voltage. Additionally, the system includes a control circuit coupled to the cathode of the one or more light emitting diodes. The control circuit is configured to receive a first voltage from the cathode of the one or more light emitting diodes, compare a second voltage and a threshold voltage, and generate a control signal based at least in part on the second voltage and the threshold voltage. The second voltage indicates a magnitude of the first voltage.

10 Claims, 5 Drawing Sheets



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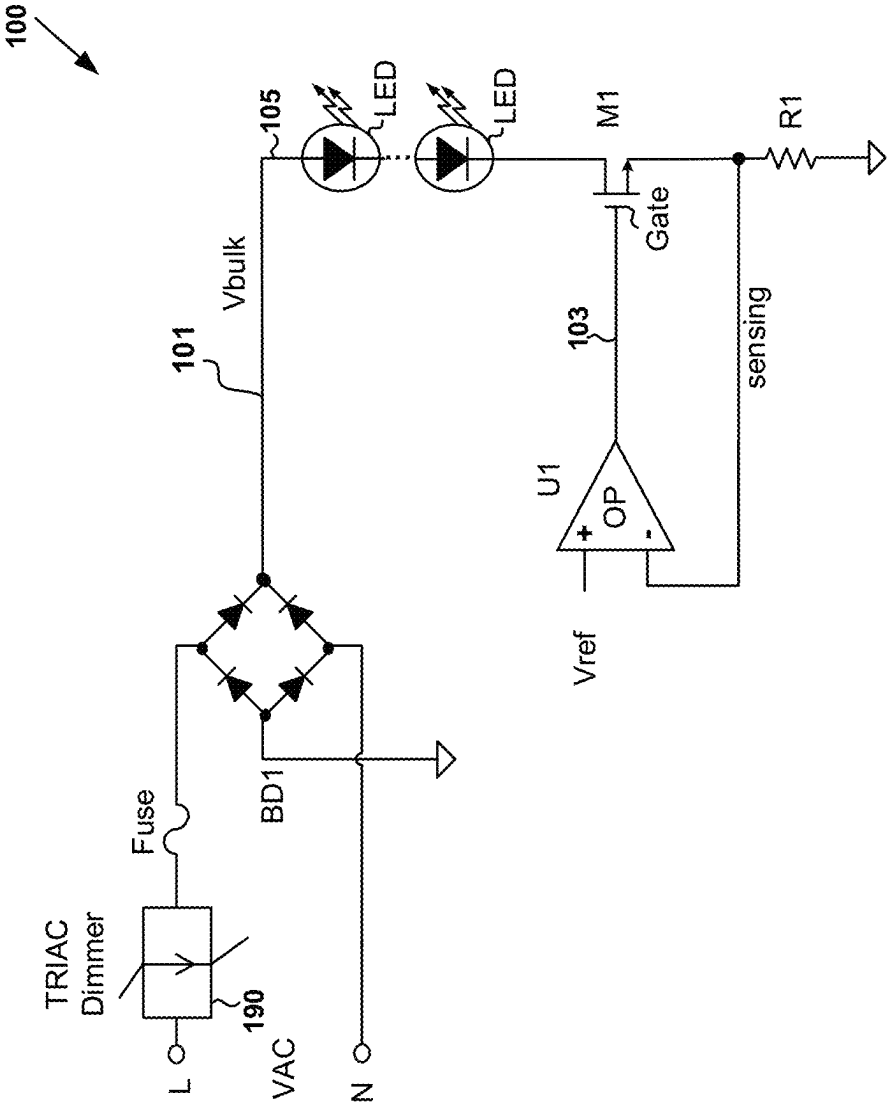


FIG. 1
PRIOR ART

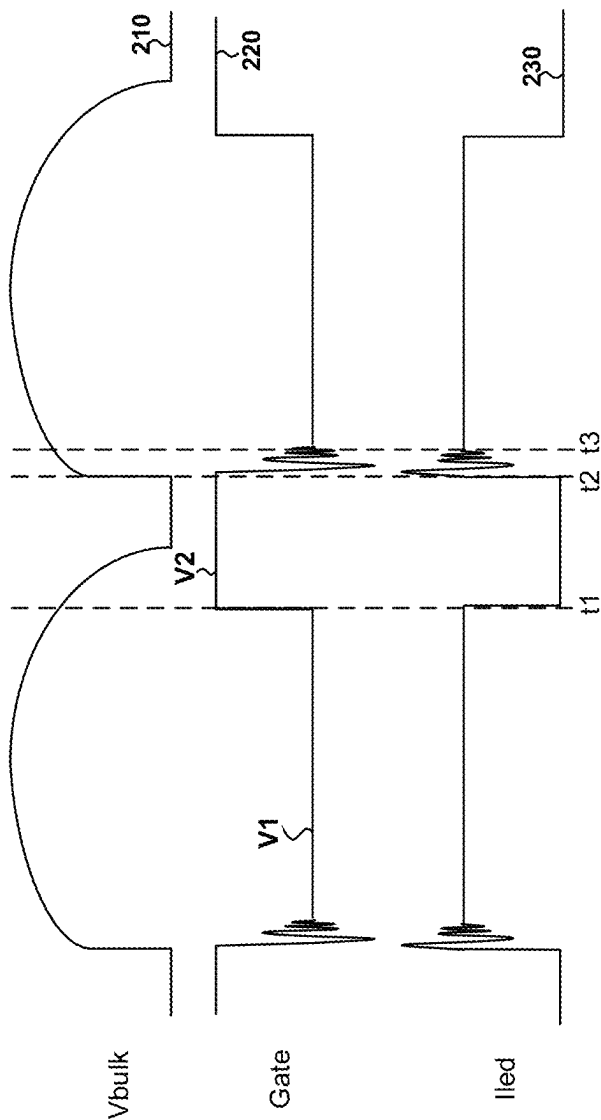


FIG. 2
PRIOR ART

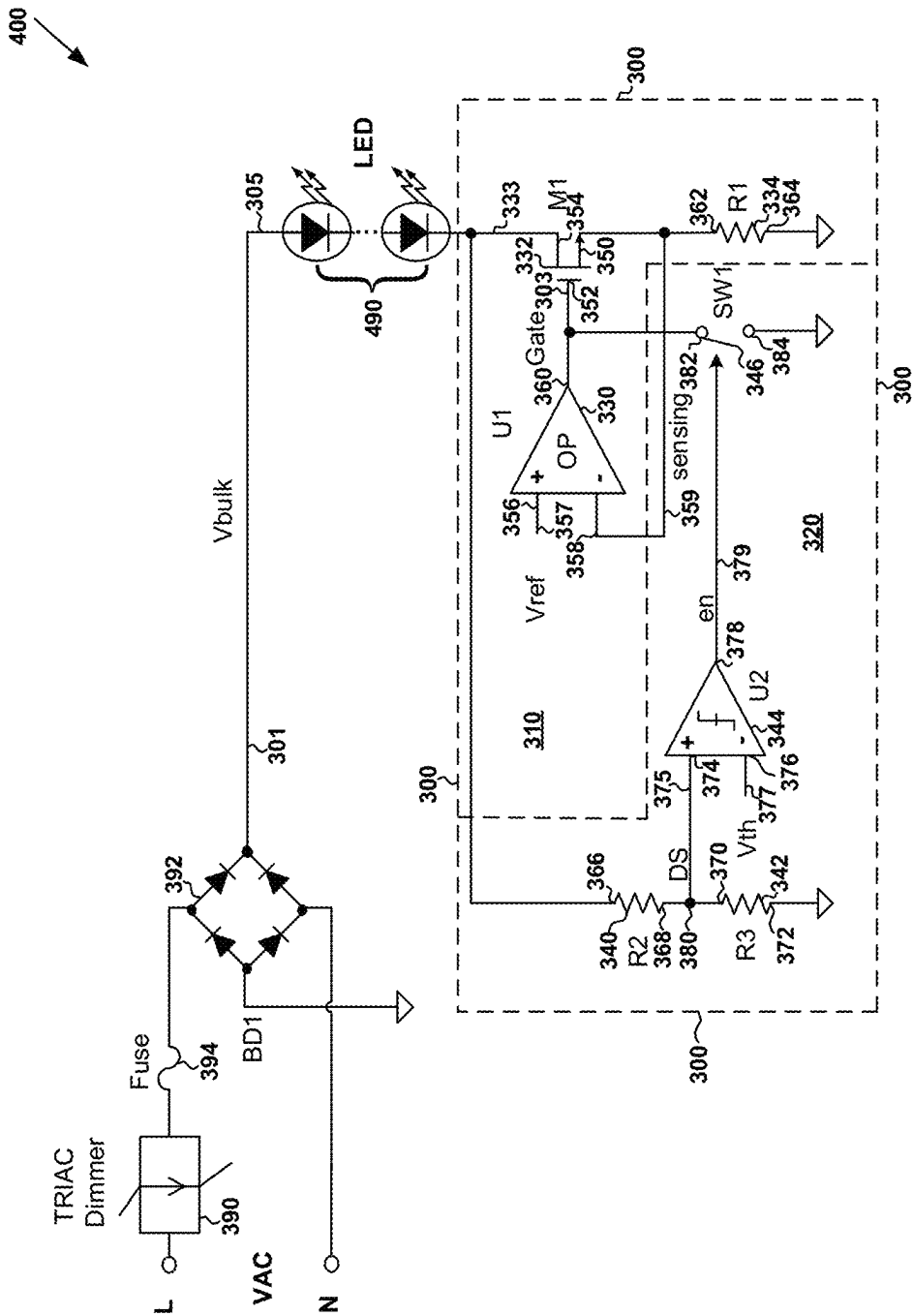


FIG. 3

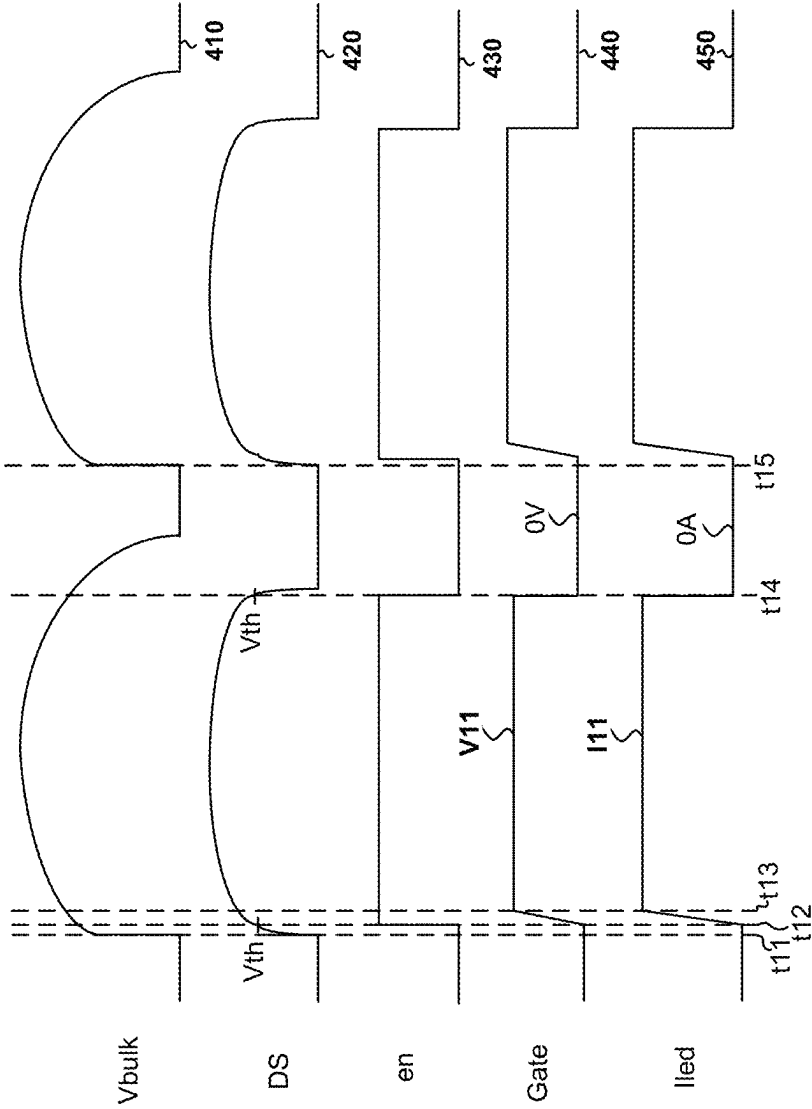


FIG. 4

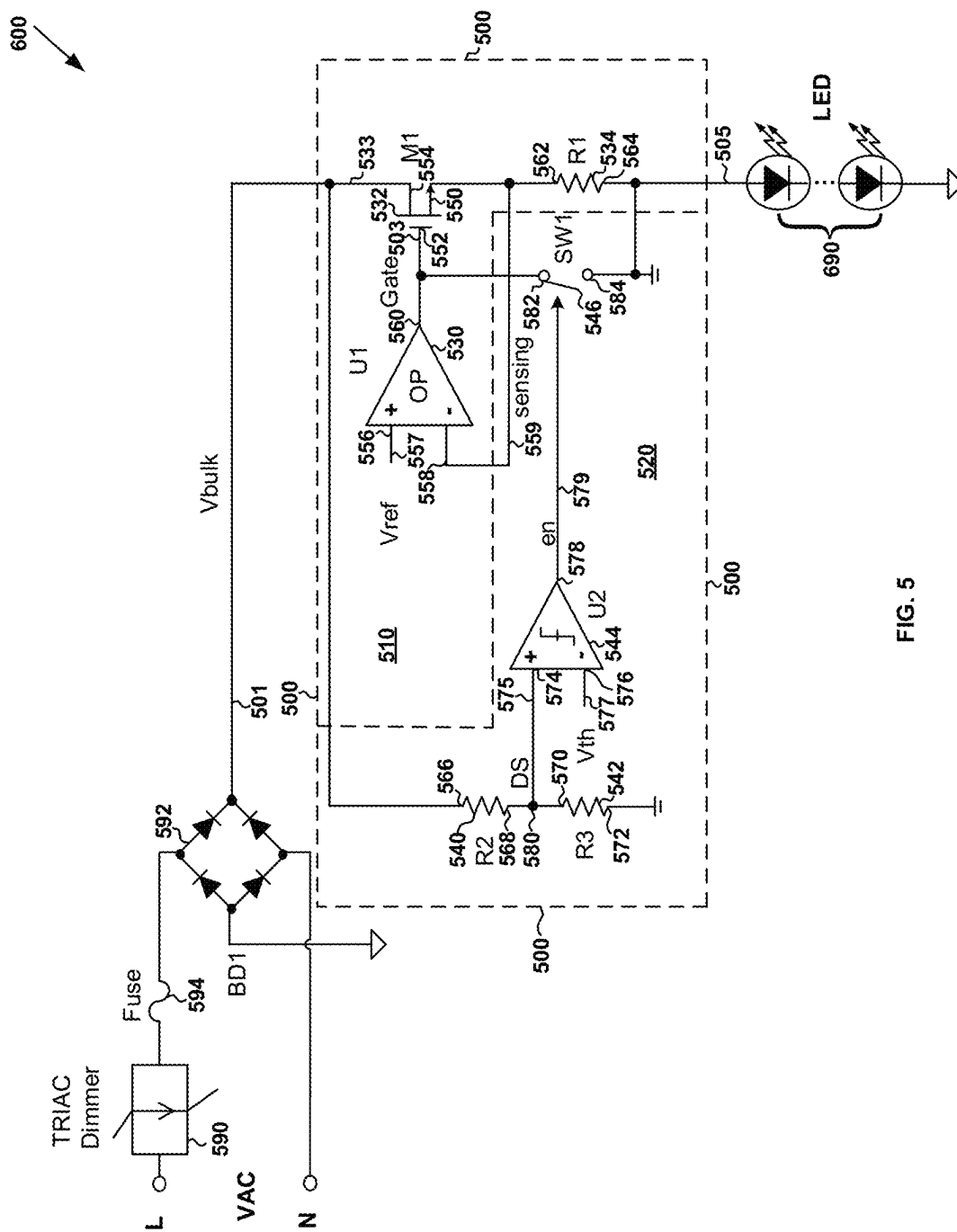


FIG. 5

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LED LIGHTING SYSTEMS WITH TRIAC DIMMERS AND METHODS THEREOF

1. CROSS-REFERENCES TO RELATED APPLICATIONS

This application is a continuation of U.S. patent application Ser. No. 18/103,971, filed Jan. 31, 2023, which is a continuation of U.S. patent application Ser. No. 17/023,632, filed Sep. 17, 2020, which is a divisional of U.S. patent application Ser. No. 16/226,424, filed Dec. 19, 2018, which claims priority to Chinese Patent Application No. 201711464007.9, filed Dec. 28, 2017, all of these applications being incorporated by reference herein for all purposes.

2. BACKGROUND OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide lighting systems and methods with Triode for Alternating Current (TRIAC) dimmers. Merely by way of example, some embodiments of the invention have been applied to light emitting diodes (LEDs). But it would be recognized that the invention has a much broader range of applicability.

As anew energy-saving and environmentally-friendly light source, light emitting diode (LED) is widely used in various fields due to its high luminance, low power consumption and long life span. For example, within a range close to a rated current, luminance of an LED often is directly proportional to the current flowing through the LED but is independent of the voltage across the LED; therefore, LED is often supplied with power from a constant current source during operation.

FIG. 1 is an exemplary circuit diagram showing a conventional linear constant current LED lighting system with a Triode for Alternating Current (TRIAC) dimmer. The linear constant current LED lighting system **100** is widely used in various fields such as LED lighting due to the system's simple and reliable structure and low cost. As shown in FIG. 1, the main control unit of the system **100** includes: an error amplifier **U1**, a transistor **M1** for power regulation, and an LED current sensing resistor **R1**. The positive input terminal of the amplifier **U1** receives a reference voltage V_{ref} and the negative input terminal of the amplifier **U1** is connected to the sensing resistor **R1**. Additionally, the output terminal of the amplifier **U1** is connected to the gate terminal of the transistor **M1** for power regulation.

As shown in FIG. 1, after the system **100** is powered on, an AC input voltage (e.g., VAC) is received by a TRIAC dimmer **190** and subjected to a full-wave rectification process performed by a full-wave rectifying bridge **BD1** to generate a rectified voltage **101** (e.g., V_{bulk}). For example, the rectified voltage **101** does not drop below 0 volt. Also, after the system **100** is powered on, the amplifier **U1** of the main control unit controls the voltage **103** of the gate terminal of the transistor **M1**, so that the transistor **M1** is closed (e.g., the transistor **M1** being turned on). As an example, the voltage **101** (e.g., V_{bulk}) is higher than a minimum forward operating voltage of the one or more LEDs, and a current **105** flows through the one or more LEDs to the sensing resistor **R1** via the transistor **M1**.

The sensing resistor **R1** includes two terminals. For example, one terminal of the sensing resistor **R1** is grounded, and another terminal of the sensing resistor **R1** generates a voltage (e.g., V_{sense}). As an example, the mag-

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nitude of the voltage (e.g., V_{sense}) across the resistor **R1** corresponds to the current **105** flowing through the one or more LEDs. The amplifier **U1** receives the voltage V_{sense} at one input terminal and receives the reference voltage V_{ref} at another input terminal, and performs an error amplification process on the voltage V_{sense} and the reference voltage V_{ref} in order to adjust the gate voltage **103** of the transistor **M1** and realize constant current control for the one or more LEDs. The LED current I_{led} (e.g., the current flowing through the one or more LEDs) is shown in Equation 1:

$$I_{led} = \frac{V_{ref}}{R_1} \quad (\text{Equation 1})$$

where R_1 represents the resistance of the resistor **R1**, and V_{ref} represents the reference voltage.

FIG. 2 shows simplified conventional timing diagrams for the LED lighting system **100** as shown in FIG. 1. The waveform **210** represents the rectified voltage V_{bulk} (e.g., the rectified voltage **101**) as a function of time, the waveform **220** represents the voltage of the gate terminal of the transistor **M1** (e.g., the gate voltage **103**) as a function of time, and the waveform **230** represents the LED current I_{led} (e.g., the current **105** flowing through the one or more LEDs) as a function of time.

In certain TRIAC dimmer applications, the AC input voltage (e.g., VAC) is, for example, clipped by the TRIAC dimmer **190** and also rectified to generate the rectified voltage **101** (e.g., V_{bulk}), which is received by an anode of the one or more LEDs. As shown by the waveform **210**, during one or more time durations (e.g., from time t_1 to time t_2), the rectified voltage **101** (e.g., V_{bulk}) either is larger than zero but still small in magnitude or is equal to zero because the TRIAC dimmer **190** operates in an off cycle, so that the voltage **101** (e.g., V_{bulk}) is lower than the minimum forward operating voltage of the one or more LEDs. For example, during the off cycle, the TRIAC dimmer **190** clips the AC input voltage (e.g., VAC) so that the rectified voltage **101** (e.g., V_{bulk}) equals zero in magnitude. As an example, from time t_1 to time t_2 , the one or more LEDs cannot be turned on due to insufficient magnitude of the voltage **101** and the LED current I_{led} is equal to zero, as shown by the waveforms **210** and **230**. At this point (e.g., at a time between time t_1 and time t_2), the constant current control circuit of FIG. 1 is still working. For example, the voltage V_{bulk} is low and no current flows through the one or more LEDs, so the voltage of the gate terminal of the transistor **M1**, which is provided by the output terminal of the **U1** amplifier, is at a high voltage level (e.g., at a voltage level V_2 from time t_1 to time t_2 as shown by the waveform **220**). As an example, as shown by the waveform **220**, prior to the time t_1 , the voltage of the gate terminal of the transistor **M1** is at a low voltage level (e.g., at a voltage level V_1). For example, the voltage level V_2 is higher than the voltage level V_1 , and the voltage level V_1 is higher than a threshold voltage of the gate terminal for turning on the transistor **M1**.

As shown in FIG. 2, at time t_2 , the TRIAC dimmer **190** enters an on cycle, during which, the TRIAC dimmer **190** does not clip the AC input voltage (e.g., VAC), as shown by the waveform **210**. For example, at time t_2 , the V_{bulk} voltage increases rapidly while the voltage of the gate terminal of the transistor **M1** is at the high voltage level, causing the LED current I_{led} to overshoot, as shown by the waveforms **210**, **220** and **230**. During the loop adjustment process for constant current of the LED lighting system **100**, after overshoot

of the LED current I_{led} , there is an oscillation period (e.g., from time t_2 to time t_3) as shown by the waveforms **220** and **230**.

As discussed above, a resulting problem of the LED lighting system **100** is that the change in the V_{bulk} voltage caused by the TRIAC dimmer **190** can cause overshoot and oscillation of the LED current I_{led} . For some TRIAC dimmers, this type of current overshoot and oscillation may even cause the TRIAC dimmer (e.g., the TRIAC dimmer **190**) to operate abnormally (e.g., causing the TRIAC dimmer **190** to misfire), which in turn causes abnormal fluctuations in the LED current I_{led} and causes the one or more LEDs to flicker.

Hence it is highly desirable to improve techniques related to LED lighting systems with TRIAC dimmers.

3. BRIEF SUMMARY OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide lighting systems and methods with Triode for Alternating Current (TRIAC) dimmers. Merely by way of example, some embodiments of the invention have been applied to light emitting diodes (LEDs). But it would be recognized that the invention has a much broader range of applicability.

Some embodiments of the present invention provide a high-compatibility TRIAC dimmer control system for an LED lighting system and a method of using such switch control system. For example, the TRIAC dimmer control system eliminates current overshoot and oscillation caused by a sudden change of a rectified voltage during a TRIAC dimming process by controlling enablement of a constant current circuit of the system. As an example, the TRIAC dimmer control system prevents a TRIAC dimmer from operating abnormally due to current surges and thus improves the compatibility of LED lighting system with TRIAC dimmer.

In certain embodiments, a system for LED switch control is provided. For example, the system includes a rectifying module configured to rectify an input voltage that has been processed by a TRIAC dimmer and to transmit the rectified voltage to a combination of one or more LEDs, the combination of the one or more LEDs being coupled to a constant current module and an enablement control module. As an example, the system also includes the enablement control module configured to receive a sensing voltage corresponding to the rectified voltage, to compare the sensing voltage with a predetermined threshold voltage, to output an enablement signal at a logic low level if the sensing voltage is lower than the predetermined threshold voltage, and to output the enablement signal at a logic high level if the sensing voltage is higher than the predetermined threshold voltage. As an example, the system also includes the constant current module configured to receive the enablement signal, to allow a current to flow through the combination of the one or more LEDs if the enablement signal is at the logic high level, and to not allow the current to flow through the combination of the one or more LEDs if the enablement signal is at the logic low level. In some embodiments, an LED lighting system including an LED switch control system is provided.

According to certain embodiments, a system for controlling one or more light emitting diodes includes a current regulation circuit coupled to a cathode of one or more light emitting diodes. The one or more light emitting diodes include the cathode and an anode configured to receive a rectified voltage. Additionally, the system includes a control

circuit coupled to the cathode of the one or more light emitting diodes. The control circuit is configured to receive a first voltage from the cathode of the one or more light emitting diodes, compare a second voltage and a threshold voltage, and generate a control signal based at least in part on the second voltage and the threshold voltage. The second voltage indicates a magnitude of the first voltage. The control circuit is further configured to: if the second voltage is larger than the threshold voltage, generate the control signal at a first logic level; and if the second voltage is smaller than the threshold voltage, generate the control signal at a second logic level. The current regulation circuit is configured to: receive the control signal from the control circuit; allow a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allow the current to flow through the one or more light emitting diodes if the control signal is at the second logic level.

According to some embodiments, a system for controlling one or more light emitting diodes includes a current regulation circuit configured to receive a rectified voltage and coupled to an anode of one or more light emitting diodes. The one or more light emitting diodes include the anode and a cathode. Additionally, the system includes a control circuit configured to receive the rectified voltage. The control circuit is configured to: compare an input voltage and a threshold voltage, the input voltage indicating a magnitude of the rectified voltage; and generate a control signal based at least in part on the input voltage and the threshold voltage. The control circuit is further configured to: if the input voltage is larger than the threshold voltage, generate the control signal at a first logic level; and if the input voltage is smaller than the threshold voltage, generate the control signal at a second logic level. The current regulation circuit is configured to: receive the control signal from the control circuit; allow a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allow the current to flow through the one or more light emitting diodes if the control signal is at the second logic level.

According to certain embodiments, a method for controlling one or more light emitting diodes includes: receiving a first voltage from a cathode of one or more light emitting diodes by a control circuit coupled to the cathode of the one or more light emitting diodes, the one or more light emitting diodes including the cathode and an anode configured to receive a rectified voltage; comparing a second voltage and a threshold voltage, the second voltage indicating a magnitude of the first voltage; generating a control signal at a first logic level if the second voltage is larger than the threshold voltage; generating the control signal at a second logic level if the second voltage is smaller than the threshold voltage; receiving the control signal from the control circuit by a current regulation circuit coupled to the cathode of one or more light emitting diodes; allowing a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allowing the current to flow through the one or more light emitting diodes if the control signal is at the second logic level.

According to some embodiments, a method for controlling one or more light emitting diodes includes: receiving a rectified voltage by a control circuit; comparing an input voltage and a threshold voltage, the input voltage indicating a magnitude of the rectified voltage; if the input voltage is

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larger than the threshold voltage, generating a control signal at a first logic level; if the input voltage is smaller than the threshold voltage, generating the control signal at a second logic level; receiving the control signal from the control circuit by a current regulation circuit configured to receive the rectified voltage and coupled to an anode of one or more light emitting diodes, the one or more light emitting diodes including the anode and a cathode; allowing a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allowing the current to flow through the one or more light emitting diodes if the control signal is at the second logic level.

Depending upon embodiment, one or more benefits may be achieved. These benefits and various additional objects, features and advantages of the present invention can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

4. BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is an exemplary circuit diagram showing a conventional linear constant current LED lighting system with a Triode for Alternating Current (TRIAC) dimmer.

FIG. 2 shows simplified conventional timing diagrams for the LED lighting system as shown in FIG. 1.

FIG. 3 is a simplified circuit diagram showing an LED lighting system with a TRIAC dimmer according to some embodiments of the present invention.

FIG. 4 shows simplified timing diagrams for controlling the LED lighting system 400 as shown in FIG. 3 according to some embodiments of the present invention.

FIG. 5 is a simplified circuit diagram showing an LED lighting system with a TRIAC dimmer according to certain embodiments of the present invention.

Depending upon embodiment, one or more benefits may be achieved. These benefits and various additional objects, features and advantages of the present invention can be fully appreciated with reference to the detailed description and accompanying drawings that follow.

5. DETAILED DESCRIPTION OF THE INVENTION

Certain embodiments of the present invention are directed to circuits. More particularly, some embodiments of the invention provide lighting systems and methods with Triode for Alternating Current (TRIAC) dimmers. Merely by way of example, some embodiments of the invention have been applied to light emitting diodes (LEDs). But it would be recognized that the invention has a much broader range of applicability.

FIG. 3 is a simplified circuit diagram showing an LED lighting system with a TRIAC dimmer according to some embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown in FIG. 3, the LED lighting system 400 includes a line (L) terminal and a neutral (N) terminal, and the system 400 also includes a TRIAC dimmer 390, a full-wave rectifying bridge 392 (e.g., a full-wave rectifying bridge BD1), a fuse 394, and a system controller 300. In some examples, the system controller 300 includes a constant current circuit 310 and a control circuit 320 (e.g., an enablement control circuit). For example, the system controller 300 is located on a chip. As an example, the constant current circuit 310

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includes an amplifier 330 (e.g., an error amplifier U1), a transistor 332 (e.g., a transistor M1 for power regulation), and a resistor 334 (e.g., a sensing resistor R1). For example, the control circuit 320 includes a resistor 340 (e.g., a resistor R2), a resistor 342 (e.g., a resistor R3), a comparator 344 (e.g., a comparator U2), and a switch 346 (e.g., a switch SW1). In certain examples, the LED lighting system 400 provides a current 305 (e.g., an LED current I_{led}) that flows through one or more LEDs 490, the transistor 332 (e.g., the transistor M1 for power regulation), and the resistor 334 (e.g., the sensing resistor R1). For example, one or more LEDs 490 include multiple LEDs connected in series. Although the above has been shown using a selected group of components for the LED lighting system, there can be many alternatives, modifications, and variations. For example, some of the components may be expanded and/or combined. Other components may be inserted to those noted above. Depending upon the embodiment, the arrangement of components may be interchanged with others replaced. Further details of these components are found throughout the present specification.

In certain embodiments, an AC input voltage (e.g., VAC) is received by the TRIAC dimmer 390 and also rectified (e.g., by the full-wave rectifying bridge 392) to generate a rectified voltage 301 (e.g., a rectified voltage V_{bulk}). For example, the full-wave rectifying bridge 392 is coupled to the TRIAC dimmer 390 through the fuse 394. As an example, the rectified voltage 301 does not fall below the ground voltage of the system (e.g., zero volt). In some embodiments, the transistor 332 (e.g., the transistor M1 for power regulation) includes a source terminal 350, a gate terminal 352 and a drain terminal 354, the amplifier 330 (e.g., the error amplifier U1) includes input terminals 356 and 358 and an output terminal 360, and the resistor 334 (e.g., the sensing resistor R1) includes terminals 362 and 364. For example, the source terminal 350 of the transistor 332 is connected to the terminal 362 of the resistor 334, the gate terminal 352 of the transistor 332 is connected to the output terminal 360 of the amplifier 330, and the drain terminal 354 of the transistor 332 is connected to a cathode of the one or more LEDs 490. As an example, an anode of the one or more LEDs 490 receives the rectified voltage 301 (e.g., a rectified voltage V_{bulk}), and the terminal 364 of the resistor 334 is biased to the ground voltage of the system (e.g., zero volt).

According to certain embodiments, the resistor 340 (e.g., the resistor R2) includes terminals 366 and 368, the resistor 342 (e.g., the resistor R3) includes terminals 370 and 372, and the comparator 344 (e.g., the comparator U2) includes an input terminal 374 (e.g., a non-inverting terminal), an input terminal 376 (e.g., an inverting terminal), and an output terminal 378. For example, the terminal 366 of the resistor 340 is connected to the drain terminal 354 of the transistor 332, and the terminal 372 of the resistor 342 is biased to the ground voltage of the system (e.g., zero volt). As an example, the terminal 368 of the resistor 340 and the terminal 370 of the resistor 342 are connected at a node 380 (e.g., a node DS), and the node 380 (e.g., the node DS) is connected to the input terminal 374 of the comparator 344. For example, the input terminal 374 of the comparator 344 is configured to detect a change of a voltage 333 of the drain terminal 354 of the transistor 332.

According to some embodiments, the input terminal 374 (e.g., the positive terminal) of the comparator 344 receives a voltage 375 of the node 380 (e.g., the node DS), which is connected to the terminal 368 of the resistor 340 and the terminal 370 of the resistor 342. For example, the voltage

375 is directly proportional to the voltage 333 of the drain terminal 354 of the transistor 332. As an example, the input terminal 376 (e.g., the negative terminal) of the comparator 344 receives a threshold voltage 377 (e.g., a threshold voltage V_{th}).

In certain embodiments, the comparator 344 compares the voltage 375 and the threshold voltage 377 and generates a control signal 379 (e.g., a control signal en). For example, if the voltage 375 is larger than the threshold voltage 377, the control signal 379 is at a logic high level. As an example, if the voltage 375 is smaller than the threshold voltage 377, the control signal 379 is at a logic low level. In some embodiments, the comparator 344 outputs the control signal 379 at the output terminal 378, and sends the control signal 379 to the switch 346 (e.g., the switch SW1) of the constant current circuit 310.

In some embodiments, the switch 346 includes terminals 382 and 384. For example, the terminal 382 is connected to the output terminal 360 of the amplifier 330, and the terminal 384 is biased to the ground voltage of the system (e.g., zero volt). As an example, the switch 346 receives the control signal 379. In certain examples, if the control signal 379 is at the logic high level, the switch 346 is open. For example, if the switch 346 is open, the constant current circuit 310 is enabled. As an example, if the switch 346 is open, a voltage 303 (e.g., a voltage V_{Gate}) of the gate terminal 352 of the transistor 332 is generated by the amplifier 330 (e.g., the error amplifier U1). In some examples, if the control signal 379 is at the logic low level, the switch 346 is closed. As an example, if the switch 346 is closed, the constant current circuit 310 is disabled. For example, if the switch 346 is closed, the voltage 303 of the gate terminal 352 of the transistor 332 is biased to the ground voltage of the system (e.g., zero volt).

According to some embodiments, if the voltage 375 is smaller than the threshold voltage 377, the constant current circuit 310 is disabled by the control signal 379. For example, if the voltage 375 is smaller than the threshold voltage 377, the voltage 333 of the drain terminal 354 of the transistor 332 is too low for the LED lighting system 400 to provide a constant current to the one or more LEDs 490. As an example, if the constant current circuit 310 is disabled, the current 305 (e.g., the LED current I_{led}) that flows through the one or more LEDs 490 is equal to zero in magnitude. For example, if the voltage 375 is smaller than the threshold voltage 377, the constant current circuit 310 does not allow the current 305 (e.g., the LED current I_{led}) with a magnitude larger than zero to flow through the one or more LEDs 490.

According to certain embodiments, if the voltage 375 is larger than the threshold voltage 377, the constant current circuit 310 is enabled by the control signal 379. As an example, if the voltage 375 is larger than the threshold voltage 377, the voltage 333 of the drain terminal 354 of the transistor 332 is high enough for the LED lighting system 400 to provide a constant current to the one or more LEDs 490. For example, if the voltage 375 is larger than the threshold voltage 377, the voltage 333 of the drain terminal 354 of the transistor 332 is higher than the minimum voltage value that is needed for the LED lighting system 400 to provide a constant current to the one or more LEDs 490. As an example, if the constant current circuit 310 is enabled, the current 305 that flows through the one or more LEDs 490 is equal to a constant that is larger than zero in magnitude. For example, if the voltage 375 is larger than the threshold voltage 377, the constant current circuit 310 allows the current 305 (e.g., the LED current I_{led}) with a magnitude larger than zero to flow through the one or more LEDs 490.

As shown in FIG. 3, the amplifier 330 includes the input terminal 356 (e.g., a non-inverting terminal), the input terminal 358 (e.g., an inverting terminal), and the output terminal 360, according to one embodiment. In some examples, the input terminal 356 (e.g., the positive terminal) receives a reference voltage 357 (e.g., a reference voltage V_{ref}). In certain examples, the input terminal 358 (e.g., the negative terminal) is connected to the source terminal 350 of the transistor 332 and the terminal 362 of the resistor 334, and receives a voltage 359 (e.g., a voltage V_{sense}). For example, the terminal 364 of the resistor 334 is biased to the ground voltage of the system (e.g., zero volt), and the voltage 359 at the terminal 362 of the resistor 334 corresponds to the current 305 that flows through the one or more LEDs 490. As an example, if the switch 346 is open, the amplifier 330 (e.g., the error amplifier U1) generates the voltage 303 based at least in part on the reference voltage 357 (e.g., the reference voltage V_{ref}) and the voltage 359 (e.g., the voltage V_{sense}), and outputs the voltage 303 at the output terminal 360. For example, the voltage 303 is received by the gate terminal 352 of the transistor 332.

In some embodiments, after the constant current circuit 310 becomes enabled by the control signal 379, the voltage 303 of the gate terminal 352 generated by the amplifier 330 (e.g., the error amplifier U1) slowly increases from the ground voltage of the system (e.g., zero volt) to a desired voltage value, and the current 305 that flows through the one or more LEDs 490 also slowly increases from zero to a desired current value. For example, the voltage 303 slowly increases from the ground voltage of the system (e.g., zero volt) to the desired voltage value without overshoot and/or oscillation. As an example, the current 305 slowly increases from zero to the desired current value without overshoot and/or oscillation.

In certain embodiments, as shown in FIG. 3, the transistor 332 (e.g., the transistor M1 for power regulation) is a field effect transistor (e.g., a metal-oxide-semiconductor field effect transistor (MOSFET)). For example, the transistor 332 is an insulated gate bipolar transistor (IGBT). As an example, the transistor 332 is a bipolar junction transistor. In some examples, the system controller 300 includes more components or less components. In certain examples, the value of the reference voltage 357 (e.g., the reference voltage V_{ref}) and/or the value of the threshold voltage 377 (e.g., the threshold voltage V_{th}) can be set as desired by those skilled in the art.

FIG. 4 shows simplified timing diagrams for controlling the LED lighting system 400 as shown in FIG. 3 according to some embodiments of the present invention. These diagrams are merely examples, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. The waveform 410 represents the rectified voltage V_{bulk} (e.g., the rectified voltage 301) as a function of time, the waveform 420 represents the voltage 375 of the node DS (e.g., the node 380) as a function of time, the waveform 430 represents the control signal en (e.g., the control signal 379) as a function of time, the waveform 440 represents the voltage V_{Gate} (e.g., the voltage 303) as a function of time, and the waveform 450 represents the LED current I_{led} (e.g., the current 305) as a function of time. According to some embodiments, the time period from time t_{11} to time t_{15} represents a half cycle of the AC input voltage (e.g., VAC). For example, the time period from time t_{11} to time t_{15} is equal to half a period of the AC input voltage (e.g., VAC).

According to certain embodiments, after the system 400 is powered on, the AC input voltage (e.g., VAC) is received

by the TRIAC dimmer 390 and subjected to a full-wave rectification process performed by the full-wave rectifying bridge 392 (e.g., the full-wave rectifying bridge BD1) to generate the rectified voltage 301 (e.g., the rectified voltage V_{bulk}). According to some embodiments, at time t_{11} , the TRIAC dimmer 390 enters an on cycle, during which, the TRIAC dimmer 390 does not clip the AC input voltage (e.g., VAC). For example, at time t_{11} , the rectified voltage 301 (e.g., the rectified voltage V_{bulk}) increases rapidly as shown by the waveform 410.

In some examples, if the rectified voltage 301 (e.g., the rectified voltage V_{bulk}) becomes larger than a magnitude that is needed to provide a minimum forward operating voltage to the one or more LEDs 490, the rectified voltage 301 (e.g., the rectified voltage V_{bulk}) causes the voltage 333 of the drain terminal 354 of the transistor 332 to increase. In certain examples, the voltage 333 is received by a voltage divider including the resistor 340 (e.g., the resistor R2) and the resistor 342 (e.g., the resistor R3), and the voltage divider generates the voltage 375. As an example, if the voltage 333 of the drain terminal 354 of the transistor 332 increases, the voltage 375 of the node 380 (e.g., the node DS) also increases.

According to certain embodiments, at time t_{11} , the TRIAC dimmer 390 stops clipping the AC input voltage (e.g., VAC), and the voltage 375 of the node 380 (e.g., the node DS) starts to increase but remains smaller than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. For example, at time t_{11} , the voltage 375 remains smaller than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic low level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 remains biased to the ground voltage of the system (e.g., zero volt) as shown by the waveform 440. As an example, at time t_{11} , the constant current circuit 310 remains disabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) remains equal to zero in magnitude as shown by the waveform 450.

According to some embodiments, from time t_{11} to time t_{12} , the voltage 375 of the node 380 (e.g., the node DS) increases but remains smaller than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. For example, from time t_{11} to time t_{12} , the voltage 375 remains smaller than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic low level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 remains biased to the ground voltage of the system (e.g., zero volt) as shown by the waveform 440. As an example, from time t_{11} to time t_{12} , the constant current circuit 310 remains disabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) remains equal to zero in magnitude as shown by the waveform 450.

According to certain embodiments, at time t_{12} , the voltage 375 of the node 380 (e.g., the node DS) becomes larger than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. For example, at time t_{12} , the voltage 375 becomes larger than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) changes from the logic low level to the logic high level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 starts to increase from the ground voltage of the system (e.g., zero volt) as shown by the

waveform 440. As an example, at time t_{12} , the constant current circuit 310 becomes enabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) starts to increase from zero in magnitude as shown by the waveform 450.

According to some embodiments, from time t_{12} to time t_{13} , the voltage 375 of the node 380 (e.g., the node DS) remains larger than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. In certain examples, from time t_{12} to time t_{13} , the voltage 375 remains larger than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic high level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 increases from the ground voltage of the system (e.g., zero volt) to a voltage level V_{11} , as shown by the waveform 440. For example, the voltage level V_{11} is higher than a threshold voltage of the gate terminal for turning on the transistor 332 (e.g., the transistor M1 for power regulation). In some examples, from time t_{12} to time t_{13} , the constant current circuit 310 remains enabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) increases from zero to a current level I_{11} as shown by the waveform 450. As an example, the current level I_{11} is a predetermined magnitude.

According to certain embodiments, at time t_{13} , the voltage 375 of the node 380 (e.g., the node DS) remains larger than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. In certain examples, at time t_{13} , the voltage 375 remains larger than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic high level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 reaches the voltage level V_{11} , as shown by the waveform 440. In some examples, at time t_{13} , the constant current circuit 310 remains enabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) reaches the current level I_{11} as shown by the waveform 450.

According to some embodiments, from time t_{13} to time t_{14} , the voltage 375 of the node 380 (e.g., the node DS) remains larger than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. In certain examples, from time t_{13} to time t_{14} , the voltage 375 remains larger than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic high level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 remains constant at the voltage level V_{11} , as shown by the waveform 440. In some examples, from time t_{13} to time t_{14} , the constant current circuit 310 remains enabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) remains constant at the current level I_{11} as shown by the waveform 450.

According to certain embodiments, at time t_{14} , the voltage 375 of the node 380 (e.g., the node DS) becomes smaller than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. For example, at time t_{14} , the voltage 375 becomes smaller than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) changes from the logic high level to the logic low level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 decreases from the voltage level V_{11} to the ground voltage of the system (e.g., zero volt) as shown by the waveform 440. As an

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example, at time t_{14} , the constant current circuit 310 becomes disabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) decreases from the current level I_{11} to zero as shown by the waveform 450.

According to some embodiments, from time t_{14} to time t_{15} , the voltage 375 of the node 380 (e.g., the node DS) remains smaller than the threshold voltage 377 (e.g., the threshold voltage V_{th}) as shown by the waveform 420. For example, from time t_{14} to time t_{15} , the voltage 375 remains smaller than the threshold voltage 377 as shown by the waveform 420, the control signal 379 (e.g., the control signal en) remains at the logic low level as shown by the waveform 430, and the voltage 303 (e.g., the voltage Gate) of the gate terminal 352 of the transistor 332 remains at the ground voltage of the system (e.g., zero volt) as shown by the waveform 440. As an example, from time t_{14} to time t_{15} , the constant current circuit 310 remains disabled by the control signal 379, and the current 305 (e.g., the LED current I_{led}) remains at zero as shown by the waveform 450.

FIG. 5 is a simplified circuit diagram showing an LED lighting system with a TRIAC dimmer according to certain embodiments of the present invention. This diagram is merely an example, which should not unduly limit the scope of the claims. One of ordinary skill in the art would recognize many variations, alternatives, and modifications. As shown in FIG. 5, the LED lighting system 600 includes a line (L) terminal and a neutral (N) terminal, and the system 600 also includes a TRIAC dimmer 590, a full-wave rectifying bridge 592 (e.g., a full-wave rectifying bridge BD1), a fuse 594, and a system controller 500. In some examples, the system controller 500 includes a constant current circuit 510 and a control circuit 520 (e.g., an enablement control circuit). For example, the system controller 500 is located on a chip. As an example, the constant current circuit 510 includes an amplifier 530 (e.g., an error amplifier U1), a transistor 532 (e.g., a transistor M1 for power regulation), and a resistor 534 (e.g., a sensing resistor R1). For example, the control circuit 520 includes a resistor 540 (e.g., a resistor R2), a resistor 542 (e.g., a resistor R3), a comparator 544 (e.g., a comparator U2), and a switch 546 (e.g., a switch SW1). In certain examples, the LED lighting system 400 provides a current 505 (e.g., an LED current I_{led}) that flows through the transistor 332 (e.g., the transistor M1 for power regulation), the resistor 334 (e.g., the sensing resistor R1), and one or more LEDs 690. For example, one or more LEDs 690 include multiple LEDs connected in series. Although the above has been shown using a selected group of components for the LED lighting system, there can be many alternatives, modifications, and variations. For example, some of the components may be expanded and/or combined. Other components may be inserted to those noted above. Depending upon the embodiment, the arrangement of components may be interchanged with others replaced. Further details of these components are found throughout the present specification.

In certain embodiments, an AC input voltage (e.g., VAC) is received by the TRIAC dimmer 590 and also rectified (e.g., by the full-wave rectifying bridge 592) to generate a rectified voltage 501 (e.g., a rectified voltage V_{bulk}). For example, the full-wave rectifying bridge 592 is coupled to the TRIAC dimmer 590 through the fuse 594. As an example, the rectified voltage 501 does not fall below the ground voltage of the system (e.g., zero volt). In some embodiments, the transistor 532 (e.g., the transistor M1 for power regulation) includes a source terminal 550, a gate terminal 552 and a drain terminal 554, the amplifier 530 (e.g., the error amplifier U1) includes input terminals 556

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and 558 and an output terminal 560, and the resistor 534 (e.g., the sensing resistor R1) includes terminals 562 and 564. For example, the source terminal 550 of the transistor 532 is connected to the terminal 562 of the resistor 534, the gate terminal 552 of the transistor 532 is connected to the output terminal 560 of the amplifier 530, and the drain terminal 554 of the transistor 532 receives a rectified voltage 501 (e.g., a rectified voltage V_{bulk}). As an example, the terminal 564 of the resistor 534 is connected to an anode of the one or more LEDs 690, and a cathode of the one or more LEDs 690 is biased to the ground voltage of the system (e.g., zero volt). For example, the terminal 564 of the resistor 534 and the anode of the one or more LEDs 690 are biased to the ground voltage of the chip (e.g., the floating ground). As an example, the ground voltage of the chip (e.g., the floating ground) is different from the ground voltage of the system (e.g., zero volt).

According to certain embodiments, the resistor 540 (e.g., the resistor R2) includes terminals 566 and 568, the resistor 542 (e.g., the resistor R3) includes terminals 570 and 572, and the comparator 544 (e.g., the comparator U2) includes an input terminal 574 (e.g., a non-inverting terminal), an input terminal 576 (e.g., an inverting terminal), and an output terminal 578. In some examples, the terminal 566 of the resistor 540 is connected to the drain terminal 554 of the transistor 532, and the terminal 572 of the resistor 542 is biased to a ground voltage of the chip (e.g., a floating ground). For example, the ground voltage of the chip (e.g., the floating ground) is different from the ground voltage of the system (e.g., zero volt). As an example, the terminal 568 of the resistor 540 and the terminal 570 of the resistor 542 are connected at a node 580 (e.g., a node DS), and the node 580 (e.g., the node DS) is connected to the input terminal 574 of the comparator 544. For example, the input terminal 574 of the comparator 544 is configured to detect a change of a voltage 533 of the drain terminal 554 of the transistor 532. As an example, the voltage 533 of the drain terminal 554 of the transistor 532 is equal to the rectified voltage 501 (e.g., the rectified voltage V_{bulk}).

According to some embodiments, the input terminal 574 (e.g., the positive terminal) of the comparator 544 receives a voltage 575 of the node 580 (e.g., the node DS), which is connected to the terminal 568 of the resistor 540 and the terminal 570 of the resistor 542. For example, the voltage 575 is directly proportional to the voltage 533 of the drain terminal 554 of the transistor 532. As an example, the input terminal 576 (e.g., the negative terminal) of the comparator 544 receives a threshold voltage 577 (e.g., a threshold voltage V_{th}).

In certain embodiments, the comparator 544 compares the voltage 575 and the threshold voltage 577 and generates a control signal 579 (e.g., a control signal en). For example, if the voltage 575 is larger than the threshold voltage 577, the control signal 579 is at a logic high level. As an example, if the voltage 575 is smaller than the threshold voltage 577, the control signal 579 is at a logic low level. In some embodiments, the comparator 544 outputs the control signal 579 at the output terminal 578, and sends the control signal 579 to the switch 546 (e.g., the switch SW1) of the constant current circuit 510.

In some embodiments, the switch 546 includes terminals 582 and 584. For example, the terminal 582 is connected to the output terminal 560 of the amplifier 530, and the terminal 584 is biased to the ground voltage of the chip (e.g., the floating ground). For example, the ground voltage of the chip (e.g., the floating ground) is different from the ground voltage of the system (e.g., zero volt). As an example, the

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switch 546 receives the control signal 579. In certain examples, if the control signal 579 is at the logic high level, the switch 546 is open. For example, if the switch 546 is open, the constant current circuit 510 is enabled. As an example, if the switch 546 is open, a voltage 503 (e.g., a voltage Gate) of the gate terminal 552 of the transistor 532 is generated by the amplifier 530 (e.g., the error amplifier U1). In some examples, if the control signal 579 is at the logic low level, the switch 546 is closed. As an example, if the switch 546 is closed, the constant current circuit 510 is disabled. For example, if the switch 546 is closed, the voltage 503 of the gate terminal 552 of the transistor 532 is biased to the ground voltage of the chip (e.g., the floating ground). As an example, the ground voltage of the chip (e.g., the floating ground) is different from the ground voltage of the system (e.g., zero volt).

According to some embodiments, if the voltage 575 is smaller than the threshold voltage 577, the constant current circuit 510 is disabled by the control signal 579. For example, if the voltage 575 is smaller than the threshold voltage 577, the voltage 533 of the drain terminal 554 of the transistor 532 is too low for the LED lighting system 600 to provide a constant current to the one or more LEDs 690. As an example, if the constant current circuit 510 is disabled, the current 505 (e.g., the LED current I_{led}) that flows through the one or more LEDs 690 is equal to zero in magnitude. For example, if the voltage 575 is smaller than the threshold voltage 577, the constant current circuit 510 does not allow the current 505 (e.g., the LED current I_{led}) with a magnitude larger than zero to flow through the one or more LEDs 690.

According to certain embodiments, if the voltage 575 is larger than the threshold voltage 577, the constant current circuit 510 is enabled by the control signal 579. As an example, if the voltage 575 is larger than the threshold voltage 577, the voltage 533 of the drain terminal 554 of the transistor 532 is high enough for the LED lighting system 600 to provide a constant current to the one or more LEDs 690. For example, if the voltage 575 is larger than the threshold voltage 577, the voltage 533 of the drain terminal 554 of the transistor 532 is higher than the minimum voltage value that is needed for the LED lighting system 600 to provide a constant current to the one or more LEDs 690. As an example, if the constant current circuit 510 is enabled, the current 505 that flows through the one or more LEDs 690 is equal to a constant that is larger than zero in magnitude. For example, if the voltage 575 is larger than the threshold voltage 577, the constant current circuit 510 allows the current 505 (e.g., the LED current I_{led}) with a magnitude larger than zero to flow through the one or more LEDs 690.

As shown in FIG. 5, the amplifier 530 includes the input terminal 556 (e.g., a non-inverting terminal), the input terminal 558 (e.g., an inverting terminal), and the output terminal 560, according to one embodiment. In some examples, the input terminal 556 (e.g., the positive terminal) receives a reference voltage 557 (e.g., a reference voltage V_{ref}). In certain examples, the input terminal 558 (e.g., the negative terminal) is connected to the source terminal 550 of the transistor 532 and the terminal 562 of the resistor 534, and receives a voltage 559 (e.g., a voltage V_{sense}). For example, the terminal 564 of the resistor 534 is connected to the anode of the one or more LEDs 690, and the voltage 559 at the terminal 562 of the resistor 534 corresponds to the current 505 that flows through the one or more LEDs 690. As an example, if the switch 546 is open, the amplifier 530 (e.g., the error amplifier U1) generates the voltage 503 based at least in part on the reference voltage 557 (e.g., the reference voltage V_{ref}) and the voltage 559 (e.g., the voltage

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V_{sense}), and outputs the voltage 503 at the output terminal 560. For example, the voltage 503 is received by the gate terminal 552 of the transistor 532.

In some embodiments, after the constant current circuit 510 becomes enabled by the control signal 579, the voltage 503 of the gate terminal 552 generated by the amplifier 530 (e.g., the error amplifier U1) slowly increases from the ground voltage of the chip (e.g., the floating ground) to a desired voltage value, and the current 505 that flows through the one or more LEDs 690 also slowly increases from zero to a desired current value. For example, the voltage 503 slowly increases from the ground voltage of the chip (e.g., the floating ground) to the desired voltage value without overshoot and/or oscillation. As an example, the current 505 slowly increases from zero to the desired current value without overshoot and/or oscillation.

In certain embodiments, as shown in FIG. 5, the transistor 532 (e.g., the transistor M1 for power regulation) is a field effect transistor (e.g., a metal-oxide-semiconductor field effect transistor (MOSFET)). For example, the transistor 532 is an insulated gate bipolar transistor (IGBT). As an example, the transistor 532 is a bipolar junction transistor. In some examples, the system controller 600 includes more components or less components. In certain examples, the value of the reference voltage 557 (e.g., the reference voltage V_{ref}) and/or the value of the threshold voltage 577 (e.g., the threshold voltage V_{th}) can be set as desired by those skilled in the art.

According to certain embodiments, at time t_{21} , the TRIAC dimmer 590 stops clipping the AC input voltage (e.g., VAC), and the voltage 575 of the node 580 (e.g., the node DS) starts to increase but remains smaller than the threshold voltage 577 (e.g., the threshold voltage V_{th}). For example, at time t_{21} , the voltage 575 remains smaller than the threshold voltage 577, the control signal 579 (e.g., the control signal en) remains at the logic low level, and the voltage 503 (e.g., the voltage Gate) of the gate terminal 552 of the transistor 532 remains biased to the ground voltage of the chip (e.g., the floating ground). As an example, at time t_{21} , the constant current circuit 510 remains disabled by the control signal 579, and the current 505 (e.g., the LED current I_{led}) remains equal to zero in magnitude.

According to some embodiments, from time t_{21} to time t_{22} , the voltage 575 of the node 580 (e.g., the node DS) increases but remains smaller than the threshold voltage 577 (e.g., the threshold voltage V_{th}). For example, from time t_{21} to time t_{22} , the voltage 575 remains smaller than the threshold voltage 577, the control signal 579 (e.g., the control signal en) remains at the logic low level, and the voltage 503 (e.g., the voltage Gate) of the gate terminal 552 of the transistor 532 remains biased to the ground voltage of the chip (e.g., the floating ground). As an example, from time t_{21} to time t_{22} , the constant current circuit 510 remains disabled by the control signal 579, and the current 505 (e.g., the LED current I_{led}) remains equal to zero in magnitude.

According to certain embodiments, at time t_{22} , the voltage 575 of the node 580 (e.g., the node DS) becomes larger than the threshold voltage 577 (e.g., the threshold voltage V_{th}). For example, at time t_{22} , the voltage 575 becomes larger than the threshold voltage 577, the control signal 579 (e.g., the control signal en) changes from the logic low level to the logic high level, and the voltage 503 (e.g., the voltage Gate) of the gate terminal 552 of the transistor 532 starts to increase from the ground voltage of the chip (e.g., the floating ground). As an example, at time t_{22} , the constant current circuit 510 becomes enabled by the control signal

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579, and the current **505** (e.g., the LED current I_{led}) starts to increase from zero in magnitude.

According to some embodiments, from time t_{22} to time t_{23} , the voltage **575** of the node **580** (e.g., the node DS) remains larger than the threshold voltage **577** (e.g., the threshold voltage V_{th}). In certain examples, from time t_{22} to time t_{23} , the voltage **575** remains larger than the threshold voltage **577**, the control signal **579** (e.g., the control signal en) remains at the logic high level, and the voltage **503** (e.g., the voltage Gate) of the gate terminal **552** of the transistor **532** increases from the ground voltage of the chip (e.g., the floating ground) to a voltage level V_{21} . For example, the voltage level V_{21} is higher than a threshold voltage of the gate terminal for turning on the transistor **532** (e.g., the transistor M1 for power regulation). In some examples, from time t_{22} to time t_{23} , the constant current circuit **510** remains enabled by the control signal **579**, and the current **505** (e.g., the LED current I_{led}) increases from zero to a current level **121**. As an example, the current level **121** is a predetermined magnitude.

According to certain embodiments, at time t_{23} , the voltage **575** of the node **580** (e.g., the node DS) remains larger than the threshold voltage **577** (e.g., the threshold voltage V_{th}). In certain examples, at time t_{23} , the voltage **575** remains larger than the threshold voltage **577**, the control signal **579** (e.g., the control signal en) remains at the logic high level, and the voltage **503** (e.g., the voltage Gate) of the gate terminal **552** of the transistor **532** reaches the voltage level V_{21} . In some examples, at time t_{23} , the constant current circuit **510** remains enabled by the control signal **579**, and the current **505** (e.g., the LED current I_{led}) reaches the current level **121**.

According to some embodiments, from time t_{23} to time t_{24} , the voltage **575** of the node **580** (e.g., the node DS) remains larger than the threshold voltage **577** (e.g., the threshold voltage V_{th}). In certain examples, from time t_{23} to time t_{24} , the voltage **575** remains larger than the threshold voltage **577**, the control signal **579** (e.g., the control signal en) remains at the logic high level, and the voltage **503** (e.g., the voltage Gate) of the gate terminal **552** of the transistor **532** remains constant at the voltage level V_{21} . In some examples, from time t_{23} to time t_{24} , the constant current circuit **510** remains enabled by the control signal **579**, and the current **505** (e.g., the LED current I_{led}) remains constant at the current level **121**.

According to certain embodiments, at time t_{24} , the voltage **575** of the node **580** (e.g., the node DS) becomes smaller than the threshold voltage **577** (e.g., the threshold voltage V_{th}). For example, at time t_{24} , the voltage **575** becomes smaller than the threshold voltage **577**, the control signal **579** (e.g., the control signal en) changes from the logic high level to the logic low level, and the voltage **503** (e.g., the voltage Gate) of the gate terminal **552** of the transistor **532** decreases from the voltage level V_{21} to the ground voltage of the chip (e.g., the floating ground). As an example, at time t_{24} , the constant current circuit **510** becomes disabled by the control signal **579**, and the current **505** (e.g., the LED current I_{led}) decreases from the current level **121** to zero.

According to some embodiments, from time t_{24} to time t_{25} , the voltage **575** of the node **580** (e.g., the node DS) remains smaller than the threshold voltage **577** (e.g., the threshold voltage V_{th}). For example, from time t_{24} to time t_{25} , the voltage **575** remains smaller than the threshold voltage **577**, the control signal **579** (e.g., the control signal en) remains at the logic low level, and the voltage **503** (e.g., the voltage Gate) of the gate terminal **552** of the transistor **532** remains at the ground voltage of the chip (e.g., the floating ground). As an example, from time t_{24} to time t_{25} ,

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the constant current circuit **510** remains disabled by the control signal **579**, and the current **505** (e.g., the LED current I_{led}) remains at zero.

In certain embodiments, a system for LED switch control is provided. For example, the system includes a rectifying module configured to rectify an input voltage that has been processed by a TRIAC dimmer and to transmit the rectified voltage to a combination of one or more LEDs, the combination of the one or more LEDs being coupled to a constant current module and an enablement control module. As an example, the system also includes the enablement control module configured to receive a sensing voltage corresponding to the rectified voltage, to compare the sensing voltage with a predetermined threshold voltage, to output an enablement signal at a logic low level if the sensing voltage is lower than the predetermined threshold voltage, and to output the enablement signal at a logic high level if the sensing voltage is higher than the predetermined threshold voltage. As an example, the system also includes the constant current module configured to receive the enablement signal, to allow a current to flow through the combination of the one or more LEDs if the enablement signal is at the logic high level, and to not allow the current to flow through the combination of the one or more LEDs if the enablement signal is at the logic low level.

In some examples, the constant current module includes an amplifier, a transistor for power regulation, and a first resistor, wherein the source of the transistor is coupled to the first resistor, the gate of the transistor is coupled to the output of the amplifier, and the drain of the transistor is coupled to a cathode of the combination of the one or more LEDs. In certain examples, the enablement control module includes a second resistor and a third resistor connected in series, and a comparator, wherein the second resistor and the third resistor are connected to the drain of the transistor at one end and grounded at another end, the non-inverting input of the comparator is connected to a connection point of the first resistor and the second resistor, and the inverting input of the comparator receives the predetermined threshold voltage.

According to certain embodiments, a system (e.g., the system controller **300**) for controlling one or more light emitting diodes includes a current regulation circuit (e.g., the constant current circuit **310**) coupled to a cathode of one or more light emitting diodes (e.g., the one or more LEDs **490**). The one or more light emitting diodes include the cathode and an anode configured to receive a rectified voltage (e.g., the rectified voltage **301**). Additionally, the system includes a control circuit (e.g., the control circuit **320**) coupled to the cathode of the one or more light emitting diodes. The control circuit is configured to receive a first voltage (e.g., the voltage **333**) from the cathode of the one or more light emitting diodes, compare a second voltage (e.g., the voltage **375**) and a threshold voltage (e.g., the threshold voltage **377**), and generate a control signal (e.g., the control signal **379**) based at least in part on the second voltage and the threshold voltage. The second voltage indicates a magnitude of the first voltage. The control circuit is further configured to: if the second voltage is larger than the threshold voltage, generate the control signal at a first logic level; and if the second voltage is smaller than the threshold voltage, generate the control signal at a second logic level. The current regulation circuit is configured to: receive the control signal from the control circuit; allow a current (e.g., the current **305**) to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allow the current to flow through the one or more light

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emitting diodes if the control signal is at the second logic level. For example, the system (e.g., the system controller 300) is implemented according to at least FIG. 3.

As an example, the second voltage is directly proportional to the first voltage in magnitude. For example, the first logic level is a logic high level, and the second logic level is a logic low level. As an example, the current regulation circuit includes: an amplifier (e.g., the amplifier 330) including a first amplifier input terminal (e.g., the input terminal 356), a second amplifier input terminal (e.g., the input terminal 358), and an amplifier output terminal (e.g., the output terminal 360); a transistor (e.g., the transistor 332) including a drain terminal (e.g., the drain terminal 354), a gate terminal (e.g., the gate terminal 352), and a source terminal (e.g., the source terminal 350); and a resistor (e.g., the resistor 334) including a first resistor terminal (e.g., the terminal 362) and a second resistor terminal (e.g., the terminal 364); wherein: the drain terminal of the transistor is coupled to the cathode of the one or more light emitting diodes; the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and the source terminal of the transistor is coupled to the first resistor terminal of the resistor and the second amplifier input terminal of the amplifier. For example, the first amplifier input terminal of the amplifier is configured to receive a reference voltage (e.g., the reference voltage 357). As an example, the second resistor terminal of the resistor is configured to receive a ground voltage (e.g., zero volt).

For example, the control circuit includes: a first resistor (e.g., the resistor 340) including a first resistor terminal (e.g., the terminal 366) and a second resistor terminal (e.g., the terminal 368); a second resistor (e.g., the resistor 342) including a third resistor terminal (e.g., the terminal 370) and a fourth resistor terminal (e.g., the terminal 372), the third resistor terminal being connected to the second resistor terminal; and a comparator (e.g., the comparator 344) including a first comparator input terminal (e.g., the input terminal 374), a second comparator input terminal (e.g., the input terminal 376), and a comparator output terminal (e.g., the output terminal 378); wherein: the first resistor terminal is coupled to the cathode of the one or more light emitting diodes and configured to receive the first voltage; and the fourth resistor terminal is configured to receive a ground voltage (e.g., zero volt); wherein: the first comparator input terminal is coupled to the second resistor terminal and the third resistor terminal and configured to receive the second voltage; the second comparator input terminal is configured to receive a threshold voltage (e.g., the threshold voltage 377); and the comparator output terminal is configured to output the control signal based at least in part on the second voltage and the threshold voltage.

As an example, the current regulation circuit includes: an amplifier (e.g., the amplifier 330) including a first amplifier input terminal (e.g., the input terminal 356), a second amplifier input terminal (e.g., the input terminal 358), and an amplifier output terminal (e.g., the output terminal 360); a transistor (e.g., the transistor 332) including a drain terminal (e.g., the drain terminal 354), a gate terminal (e.g., the gate terminal 352), and a source terminal (e.g., the source terminal 350); and a third resistor (e.g., the resistor 334) including a fifth resistor terminal (e.g., the terminal 362) and a sixth resistor terminal (e.g., the terminal 364); wherein: the drain terminal of the transistor is coupled to the cathode of the one or more light emitting diodes; the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and the source terminal of the transistor is coupled to the fifth resistor terminal of the third resistor and

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the second amplifier input terminal of the amplifier. For example, the sixth resistor terminal is configured to receive the ground voltage.

As an example, the control circuit further includes: a switch (e.g., the switch 346) including a first switch terminal (e.g., the terminal 382) and a second switch terminal (e.g., the terminal 384); wherein: the first switch terminal is connected to the amplifier output terminal (e.g., the output terminal 360) of an amplifier (e.g., the amplifier 330); and the second switch terminal is configured to receive the ground voltage. For example, the control circuit is further configured to: open the switch if the control signal is at the first logic level; and close the switch if the control signal is at the second logic level.

In some examples, from a first time (e.g., time t_{in}) to a second time (e.g., time t_{12}), the second voltage (e.g., the voltage 375) increases but remains smaller than the threshold voltage (e.g., the threshold voltage 377), and the control signal remains at the second logic level; at the second time, the second voltage (e.g., the voltage 375) becomes larger than the threshold voltage (e.g., the threshold voltage 377), and the control signal changes from the second logic level to the first logic level; from the second time (e.g., time t_{12}) to a third time (e.g., time t_{13}), the second voltage remains larger than the threshold voltage, and the control signal remains at the first logic level; at the third time (e.g., time t_{13}), the second voltage remains larger than the threshold voltage, the control signal remains at the first logic level, and the current (e.g., the current 305) reaches a predetermined current level (e.g., the current level I_{11}); from the third time (e.g., time t_{13}) to a fourth time (e.g., time t_{14}), the second voltage remains larger than the threshold voltage, the control signal remains at the first logic level, and the current (e.g., the current 305) remains constant at the predetermined current level (e.g., the current level I_{11}); at the fourth time (e.g., time t_{14}), the second voltage becomes smaller than the threshold voltage, and the control signal changes from the first logic level to the second voltage level; and from the fourth time (e.g., time t_{14}) to a fifth time (e.g., time t_{15}), the second voltage remains smaller than the threshold voltage, and the control signal remains at the second voltage level.

According to some embodiments, a system (e.g., the system controller 500) for controlling one or more light emitting diodes includes a current regulation circuit (e.g., the constant current circuit 510) configured to receive a rectified voltage (e.g., the rectified voltage 501) and coupled to an anode of one or more light emitting diodes (e.g., the one or more LEDs 690). The one or more light emitting diodes include the anode and a cathode. Additionally, the system includes a control circuit (e.g., the control circuit 520) configured to receive the rectified voltage (e.g., the rectified voltage 501). The control circuit is configured to: compare an input voltage (e.g., the voltage 575) and a threshold voltage (e.g., the threshold voltage 577), the input voltage indicating a magnitude of the rectified voltage; and generate a control signal (e.g., the control signal 579) based at least in part on the input voltage and the threshold voltage. The control circuit is further configured to: if the input voltage is larger than the threshold voltage, generate the control signal at a first logic level; and if the input voltage is smaller than the threshold voltage, generate the control signal at a second logic level. The current regulation circuit is configured to: receive the control signal from the control circuit; allow a current (e.g., the current 505) to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allow the current to flow through the one

or more light emitting diodes if the control signal is at the second logic level. For example, the system (e.g., the system controller 500) is implemented according to at least FIG. 5.

As an example, the input voltage is directly proportional to the rectified voltage in magnitude. For example, the first logic level is a logic high level; and the second logic level is a logic low level. As an example, the current regulation circuit includes: an amplifier (e.g., the amplifier 530) including a first amplifier input terminal (e.g., the input terminal 556), a second amplifier input terminal (e.g., the input terminal 558), and an amplifier output terminal (e.g., the output terminal 560); a transistor (e.g., the transistor 532) including a drain terminal (e.g., the drain terminal 554), a gate terminal (e.g., the gate terminal 552), and a source terminal (e.g., the source terminal 550); and a resistor (e.g., the resistor 534) including a first resistor terminal (e.g., the terminal 562) and a second resistor terminal (e.g., the terminal 564); wherein: the drain terminal of the transistor is configured to receive the rectified voltage (e.g., the rectified voltage 501); the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and the source terminal of the transistor is coupled to the anode of the one or more light emitting diodes (e.g., the one or more LEDs 690). For example, the first amplifier input terminal of the amplifier is configured to receive a reference voltage (e.g., the reference voltage 557). As an example, the cathode of the one or more light emitting diodes is configured to receive a ground voltage (e.g., zero volt).

For example, the control circuit includes: a first resistor (e.g., the resistor 540) including a first resistor terminal (e.g., the terminal 566) and a second resistor terminal (e.g., the terminal 568); a second resistor (e.g., the resistor 542) including a third resistor terminal (e.g., the terminal 570) and a fourth resistor terminal (e.g., the terminal 572), the third resistor terminal being connected to the second resistor terminal; and a comparator (e.g., the comparator 544) including a first comparator input terminal (e.g., the input terminal 574), a second comparator input terminal (e.g., the input terminal 576), and a comparator output terminal (e.g., the output terminal 578); wherein: the first resistor terminal is configured to receive the rectified voltage (e.g., the rectified voltage 501); and the fourth resistor terminal is configured to receive a ground voltage (e.g., the floating ground); wherein: the first comparator input terminal is coupled to the second resistor terminal and the third resistor terminal and configured to receive the input voltage; the second comparator input terminal is configured to receive a threshold voltage (e.g., the threshold voltage 577); and the comparator output terminal is configured to output the control signal based at least in part on the input voltage and the threshold voltage.

As an example, the current regulation circuit includes: an amplifier (e.g., the amplifier 530) including a first amplifier input terminal (e.g., the input terminal 556), a second amplifier input terminal (e.g., the input terminal 558), and an amplifier output terminal (e.g., the output terminal 560); a transistor (e.g., the transistor 532) including a drain terminal (e.g., the drain terminal 554), a gate terminal (e.g., the gate terminal 552), and a source terminal (e.g., the source terminal 550); and a third resistor (e.g., the resistor 534) including a fifth resistor terminal (e.g., the terminal 562) and a sixth resistor terminal (e.g., the terminal 564); wherein: the drain terminal of the transistor is configured to receive the rectified voltage (e.g., the rectified voltage 501); the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and the source terminal of the transistor is coupled to the fifth resistor terminal of the third

resistor and the second amplifier input terminal of the amplifier. For example, the sixth resistor terminal is coupled to the anode of the one or more light emitting diodes.

As an example, the control circuit further includes: a switch (e.g., the switch 546) including a first switch terminal (e.g., the terminal 582) and a second switch terminal (e.g., the terminal 584); wherein: the first switch terminal is connected to the amplifier output terminal (e.g., the output terminal 560) of an amplifier (e.g., the amplifier 530); and the second switch terminal is configured to receive the ground voltage. For example, the control circuit is further configured to: open the switch if the control signal is at the first logic level; and close the switch if the control signal is at the second logic level.

In certain examples, from a first time (e.g., time t_{21}) to a second time (e.g., time t_{22}), the second voltage (e.g., the voltage 575) increases but remains smaller than the threshold voltage (e.g., the threshold voltage 577), and the control signal remains at the second logic level; at the second time, the second voltage (e.g., the voltage 575) becomes larger than the threshold voltage (e.g., the threshold voltage 577), and the control signal changes from the second logic level to the first logic level; from the second time (e.g., time t_{22}) to a third time (e.g., time t_{23}), the second voltage remains larger than the threshold voltage, and the control signal remains at the first logic level; at the third time (e.g., time t_{23}), the second voltage remains larger than the threshold voltage, the control signal remains at the first logic level, and the current (e.g., the current 505) reaches a predetermined current level (e.g., the current level I_{21}); from the third time (e.g., time t_{23}) to a fourth time (e.g., time t_{24}), the second voltage remains larger than the threshold voltage, the control signal remains at the first logic level, and the current (e.g., the current 505) remains constant at the predetermined current level (e.g., the current level I_{21}); at the fourth time (e.g., time t_{24}), the second voltage becomes smaller than the threshold voltage, and the control signal changes from the first logic level to the second voltage level; and from the fourth time (e.g., time t_{24}) to a fifth time (e.g., time t_{25}), the second voltage remains smaller than the threshold voltage, and the control signal remains at the second voltage level.

According to certain embodiments, a method for controlling one or more light emitting diodes includes: receiving a first voltage (e.g., the voltage 333) from a cathode of one or more light emitting diodes by a control circuit (e.g., the control circuit 320) coupled to the cathode of the one or more light emitting diodes, the one or more light emitting diodes including the cathode and an anode configured to receive a rectified voltage (e.g., the rectified voltage 301); comparing a second voltage (e.g., the voltage 375) and a threshold voltage (e.g., the threshold voltage 377), the second voltage indicating a magnitude of the first voltage; generating a control signal (e.g., the control signal 379) at a first logic level if the second voltage is larger than the threshold voltage; generating the control signal at a second logic level if the second voltage is smaller than the threshold voltage; receiving the control signal from the control circuit by a current regulation circuit (e.g., the constant current circuit 310) coupled to the cathode of one or more light emitting diodes (e.g., the one or more LEDs 490); allowing a current (e.g., the current 305) to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allowing the current to flow through the one or more light emitting diodes if the control signal is at the second logic level. For example, the method is implemented according to at least FIG. 3.

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According to some embodiments, a method for controlling one or more light emitting diodes includes: receiving a rectified voltage (e.g., the rectified voltage **501**) by a control circuit (e.g., the control circuit **520**); comparing an input voltage (e.g., the voltage **575**) and a threshold voltage (e.g., the threshold voltage **577**), the input voltage indicating a magnitude of the rectified voltage; if the input voltage is larger than the threshold voltage, generating a control signal (e.g., the control signal **579**) at a first logic level; if the input voltage is smaller than the threshold voltage, generating the control signal at a second logic level; receiving the control signal from the control circuit by a current regulation circuit (e.g., the constant current circuit **510**) configured to receive the rectified voltage (e.g., the rectified voltage **501**) and coupled to an anode of one or more light emitting diodes (e.g., the one or more LEDs **690**), the one or more light emitting diodes including the anode and a cathode; allowing a current (e.g., the current **505**) to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and not allowing the current to flow through the one or more light emitting diodes if the control signal is at the second logic level. For example, the method is implemented according to at least FIG. 5.

For example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented using one or more software components, one or more hardware components, and/or one or more combinations of software and hardware components. As an example, some or all components of various embodiments of the present invention each are, individually and/or in combination with at least another component, implemented in one or more circuits, such as one or more analog circuits and/or one or more digital circuits. For example, various embodiments and/or examples of the present invention can be combined.

Although specific embodiments of the present invention have been described, it will be understood by those of skill in the art that there are other embodiments that are equivalent to the described embodiments. Accordingly, it is to be understood that the invention is not to be limited by the specific illustrated embodiments, but only by the scope of the appended claims.

What is claimed is:

1. A system for controlling one or more light emitting diodes, the system comprising:
 - a current regulation circuit coupled directly to a rectifier to receive a rectified voltage and coupled to an anode of one or more light emitting diodes, the one or more light emitting diodes including the anode and a cathode; and
 - a control circuit configured to receive the rectified voltage;
 wherein the control circuit is configured to:
 - compare an input voltage and a threshold voltage, the input voltage indicating a magnitude of the rectified voltage; and
 - generate a control signal based at least in part on the input voltage and the threshold voltage;
 wherein the control circuit is further configured to:
 - if the input voltage is larger than the threshold voltage, generate the control signal at a first logic level; and
 - if the input voltage is smaller than the threshold voltage, generate the control signal at a second logic level;
 wherein the current regulation circuit is configured to:
 - receive the control signal from the control circuit;

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- allow a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and
 - not allow the current to flow through the one or more light emitting diodes if the control signal is at the second logic level.
2. The system of claim 1 wherein the input voltage is directly proportional to the rectified voltage in magnitude.
 3. The system of claim 1 wherein:
 - the first logic level is a logic high level; and
 - the second logic level is a logic low level.
 4. The system of claim 1 wherein the control circuit includes:
 - a first resistor including a first resistor terminal and a second resistor terminal;
 - a second resistor including a third resistor terminal and a fourth resistor terminal, the third resistor terminal being connected to the second resistor terminal; and
 - a comparator including a first comparator input terminal, a second comparator input terminal, and a comparator output terminal;
 wherein:
 - the first resistor terminal is configured to receive the rectified voltage; and
 - the fourth resistor terminal is configured to receive a ground voltage;
 wherein:
 - the first comparator input terminal is coupled to the second resistor terminal and the third resistor terminal and configured to receive the input voltage;
 - the second comparator input terminal is configured to receive the threshold voltage; and
 - the comparator output terminal is configured to output the control signal based at least in part on the input voltage and the threshold voltage.
 5. The system of claim 4 wherein the current regulation circuit includes:
 - an amplifier including a first amplifier input terminal, a second amplifier input terminal, and an amplifier output terminal;
 - a transistor including a drain terminal, a gate terminal, and a source terminal; and
 - a third resistor including a fifth resistor terminal and a sixth resistor terminal;
 wherein:
 - the drain terminal of the transistor is configured to receive the rectified voltage;
 - the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and
 - the source terminal of the transistor is coupled to the fifth resistor terminal of the third resistor and the second amplifier input terminal of the amplifier.
 6. The system of claim 5 wherein the sixth resistor terminal is coupled to the anode of the one or more light emitting diodes.
 7. The system of claim 5 wherein the control circuit further includes:
 - a switch including a first switch terminal and a second switch terminal;
 wherein:
 - the first switch terminal is connected to the amplifier output terminal of the amplifier; and
 - the second switch terminal is configured to receive the ground voltage.
 8. The system of claim 7 wherein the control circuit is further configured to:

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open the switch if the control signal is at the first logic level; and
close the switch if the control signal is at the second logic level.

9. A method for controlling one or more light emitting diodes, the method comprising: 5
receiving a rectified voltage by a control circuit coupled directly to a rectifier;
comparing an input voltage and a threshold voltage, the input voltage indicating a magnitude of the rectified voltage; 10
if the input voltage is larger than the threshold voltage, generating a control signal at a first logic level;
if the input voltage is smaller than the threshold voltage, 15
generating the control signal at a second logic level;
receiving the control signal from the control circuit by a current regulation circuit configured to receive the rectified voltage and coupled to an anode of one or more light emitting diodes, the one or more light emitting diodes including the anode and a cathode; 20
allowing a current to flow through the one or more light emitting diodes if the control signal is at the first logic level, the current being larger than zero in magnitude; and
not allowing the current to flow through the one or more light emitting diodes if the control signal is at the second logic level. 25

10. A system for controlling one or more light emitting diodes, the system comprising:

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a current regulation circuit configured to receive a rectified voltage and coupled to an anode of one or more light emitting diodes, the one or more light emitting diodes including the anode and a cathode; and
a control circuit configured to receive the rectified voltage and generate a control signal based at least in part on the rectified voltage;
wherein the current regulation circuit is configured to:
receive the control signal from the control circuit;
allow a current to flow through the one or more light emitting diodes if the control signal is at a first logic level, the current being larger than zero in magnitude; and
not allow the current to flow through the one or more light emitting diodes if the control signal is at a second logic level;
wherein the current regulation circuit includes:
an amplifier including a first amplifier input terminal, a second amplifier input terminal, and an amplifier output terminal; and
a transistor including a drain terminal, a gate terminal, and a source terminal;
wherein:
the drain terminal of the transistor is configured to receive the rectified voltage;
the gate terminal of the transistor is coupled to the amplifier output terminal of the amplifier; and
the source terminal of the transistor is coupled to the anode of the one or more light emitting diodes.

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